

1 Basics/Revision

- $i = \sqrt{-1}$ is called the imaginary/complex number
- $\mathbb{C}$ is the set of all numbers of the form $z = x + yi$, i.e $\mathbb{C} = \{z : z = x + yi, x, y \in \mathbb{R}\}$
 - x is the real part of the complex number, i.e $\Re(z) = x$
 - y is the imaginary part of the complex number, i.e $\Im(z) = y$
- The magnitude/norm over complex numbers $z = x + yi$ is defined as $|z| = \sqrt{x^2 + y^2}$
- Conjugate of a complex number $z = x + yi$ is defined as $\bar{z} = x - yi$
 - $z \cdot \bar{z} = |z|^2$
- $\arg(z) = \theta \in (-\pi, \pi]$ where θ is the angle between the positive real axis and the line segment joining the origin to the point (x, y) in the complex plane.
- **adding and subtracting complex numbers**
- **multiplying complex numbers**
- **dividing complex numbers**
- Geometric interpretation of $|z_1 - z_2|$?
- **Polar form** of a complex number $z = x + yi$ is given by $z = r \operatorname{cis}(\theta) = r(\cos \theta + i \sin \theta) = \mathbf{re}^{i\theta}$ where $r = |z|$ and $\theta = \arg(z)$

- Multiplying/dividing with complex numbers is easier in polar form
- **Mathematica**
 - `Abs[z]` gives the magnitude
 - `Arg[z]` gives the argument
 - Can also use `ArcTan[x,y]!!`
 - `Conjugate[z]` gives the conjugate
 - `Re[z]` , `Im[z]` gives the real and imaginary parts
 - `ComplexExpand[z]` gives the cartesian form of a complex number
 - Will send mathematica code for more functions :)

1. For $z_1 = 3 + 4i$, find $|z_1|$, $\overline{z_1}$, $\arg(z_1)$ and express z_1 in polar form.

2. Let $z_2 = 2cis(\pi/4)$. Find z_2 in cartesian form.

3. Find the product and quotient of z_1 and z_2 in both cartesian and polar forms.

- **De Moivre's Theorem** for any complex number $z = rcis(\theta)$ and any integer n , we have $z^n = r^n cis(n\theta)$: Use to calculate powers of complex numbers
- **Roots of complex numbers** If $z = rcis(\theta)$, then the n -th roots of z are given by $z_k = r^{1/n} cis\left(\frac{\theta + 2k\pi}{n}\right)$ for $k = 0, 1, \dots, n-1$
- Geometric interpretation of the roots of a complex number?

2 Complex polynomials

- $\mathbb{C}$ is algebraically closed, i.e every non-constant polynomial with complex coefficients has at least one complex root.
- This means that a polynomial of degree n has exactly n roots in $\mathbb{C}$ (counting multiplicities)! (How?) i.e **Fundamental theorem of algebra**
- **Conjugate root theorem** If a polynomial has real coefficients and a complex root z , then its conjugate $\bar{z}$ is also a root.
- Vieta's formulas for a polynomial of degree 3 with roots r_1, r_2, r_3 and leading coefficient a_3 :

1. $r_1 + r_2 + r_3 = -\frac{a_2}{a_3}$

2. $r_1r_2 + r_1r_3 + r_2r_3 = \frac{a_1}{a_3}$

3. $r_1r_2r_3 = (-1)^3 \frac{a_0}{a_3}$

- **Conjugate Root theorem:** If a polynomial has real coefficients and a complex root z , then its conjugate $\bar{z}$ is also a root.

Let $z \in \mathbb{C}$

1. $z^4 - 2z^2 + 4 = 0$

2. $z^4 - 4z^3 + 7z^2 - 4z + 6 = 0$ given $z = i$ is a solution

3. $z^3 - (2 - i)z^2 = 2 - i - z$

4. $z^3 + 2iz^2 - 2z - i = 0$ given that $z = -i$ is a solution

5. $z^2 = i\bar{z}$, give your answer(s) in cartesian form.
6. (By hand) $z^4 - 1 - \sqrt{3}i = 0$, give your answer(s) in cartesian form where possible
7. Find (in cartesian form) all quartic roots of $z = i$
8. if $|z| = 1$ and $\text{Arg}(z) = \theta$, show that $\frac{2}{1+z} = 1 - i \tan(\theta/2)$
9. Factorise $z^3 + iz^2 + 2i$
10. write $\left(\frac{-\sqrt{3}+i}{2-2\sqrt{3}i}\right)^3$ in the form of $a + bi, a, b \in \mathbb{R}$

11. Find all solutions to $z^4 = \sqrt{3}z^2 - 1$ in polar form